Cransfield Materials Testing Center

Quality Assurance

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PARTICLE SIZE DISTRIBUTION
BS 1377 2:1990: 9.2/9.3/9.4 & 9.5

Client:	AHP					**Date submitted:**	09-Jun-26
Project/Site Name:	KIRIAINI AHP					**Date tested:**	11-Jun-26
Sample ID:	<0.063	**Sample No.:**	BH01	**Sample depth (M):**	0.0 - 25.0	**Sampled and Submitted by:**	Cransfield

Sample preparation details		
Initial quartered and air dried mass	(gm)	328
Washed and oven dried mass	(gm)	176.4
Fine mass	(gm)	151.6
Fines percent	(%)	46.2

Moisture content at sample preparations		
Wet weight of sample	(gm)	186.5
Dry weight of sample	(gm)	149.0
Wet weight of water	(gm)	37.5
Moisture content	(%)	25.2

Soil Classification

GRAVEL (%):	22.8	**CLAY/SILT (%):**	71.9	**USCS**	SILT OF HIGH PLASTICITY(MH)
SAND (%):	5.3			**AASHTO**	A-7-5

Wet & Dry Sieve Analysis to BS 1377 2:1990: 9.2/9.3/9.4

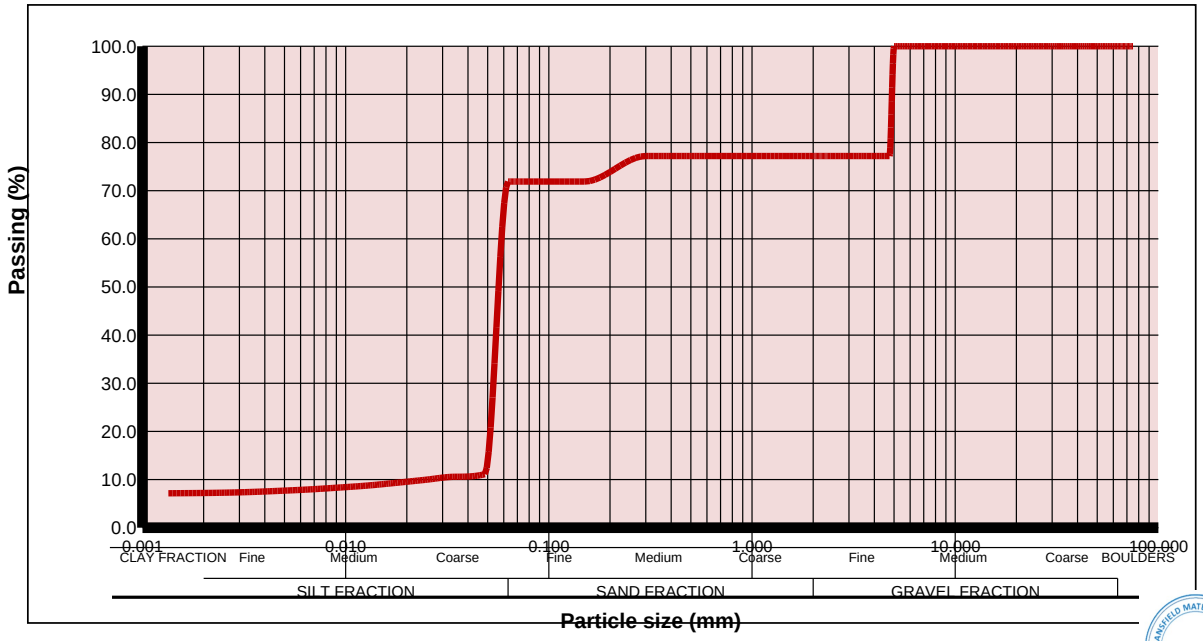
Sieve size (mm)	Retained mass (gm)	% Retained (%)	Cumulative passing percentage (%)
75	0	0.0	100.0
63	0	0.0	100.0
50	0	0.0	100.0
37.5	0	0.0	100.0
28	0	0.0	100.0
20	0	0.0	100.0
14	0	0.0	100.0
10	0	0.0	100.0
6.3	0	0.0	100.0
5	0	0.0	100.0
4.75 (No. 4)	40.22	22.8	77.2
3.35	0	0.0	77.2
2.36	0	0.0	77.2
2.00 (No. 10)	0	0.0	77.2
1.18	0	0.0	77.2
0.6	0	0.0	77.2
0.425	0	0.0	77.2
0.3	0	0.0	77.2
0.15	9.35	5.3	71.9
0.075	0	0.0	71.9
0.063	0	0.0	71.9
<0.063	126.83	71.9	-
TOTAL	176.4	-	-

Hydrometer Analysis to BS 1377-2:1990:9.5

Dry Weight	(g)	176.4	**Hydrometer Type** 152H
S.G (Mg/m³)		2.65	**Zero Correction factor** 0
Temperature (°C)		20	**Meniscus correction (g/L)** 0.1
K Factor		1.358	**Temperature Correction Factor** _____

Time, min	Actual rh	Adjusted HR(Rh)	Composite correction	Corrected HR	Effective depth (cm)	Diameter; D K√(L/T)	% Fines in Suspension	% Fines based on whole sample
0.25	24.5	24.5	0.10	24.6	12.26	0.0956	22.4	12.0
0.50	-	23.5	0.10	23.6	12.42	0.0680	21.5	11.6
1	22.6	22.6	0.10	22.7	12.57	0.0484	20.7	11.1
2	21.5	21.5	0.10	21.6	12.75	0.0345	19.7	10.6
4	20.2	20.2	0.10	20.3	12.97	0.0246	18.5	9.9
8	18.9	18.9	0.10	19.0	13.18	0.0175	17.3	9.3
15	17.8	17.8	0.10	17.9	13.36	0.0129	16.3	8.8
30	16.9	16.9	0.10	17.0	13.51	0.0092	15.5	8.3
60	16.1	16.1	0.10	16.2	13.64	0.0065	14.7	7.9
120	15.5	15.5	0.10	15.6	13.74	0.0046	14.2	7.6
240	15.0	15.0	0.10	15.1	13.82	0.0033	13.7	7.4
480	14.7	14.7	0.10	14.8	13.87	0.0023	13.5	7.2
1440	14.5	14.5	0.10	14.6	13.90	0.0013	13.3	7.1

PARTICLE SIZE DISTRIBUTION GRAPH



Tested by:	JILLO	**Date reported:**	_____	Checked by: _____